

CHER SCARLETT

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EDUCATION

2028 WELLESLEY COLLEGE

May B.A., Astrophysics; Minor in Education Studies (*expected*)

Extracurriculars: A.S.T.R.O., Society of Physics Students (SPS)

Includes cross-enrollment at the Massachusetts Institute of Technology (MIT).

2026 ORANGE COAST COLLEGE

May A.S., Physics, Mathematics, Astronomy, & Natural Science; A.A., Liberal Arts

Honors: Graduated *Summa Cum Laude*; President's List

Extracurriculars: Honors Program, Astronomy Club (*Treasurer*), STEM Club (*Outreach Manager*), MESA, EOPS, Math & Science Club (*President*), SPS, $\Phi\Theta\K$, $\Lambda\Gamma\Sigma$, $\Sigma\Lambda\Pi$

Includes enrollment in other California community colleges.

SKILLS & PROFESSIONAL ASSOCIATIONS

Python | Java | Groovy | Ruby | PHP | C | JavaScript | CSS | HTML | IDL | Fortran | Digistar
LaTeX | Mathematica | MATLAB | Jupyter Notebooks | AstroImageJ | ArcGIS | PyTorch

Institute of Physics, American Physical Society, American Geophysical Union, American
Mathematical Society, American Association of Physics Teachers, American Meteorological
Society, Simulating eXtreme Spacetimes (SXS) group

RESEARCH EXPERIENCES

2026 UNIVERSITY OF CALIFORNIA, BERKELEY

Space Sciences Lab

Detecting Atmospheric Gravity Waves with Far Ultraviolet Emissions

American Institute of Physics (AIP) TUT Fellow | Advisor: Dr. L. Claire Gasque

2026 CUESTA COLLEGE

Double Star Astrometry | Advisor: Dr. James Eickemeyer

2025 INSTITUTE FOR STUDENT ASTRONOMICAL RESEARCH (INSTAR)

Las Cumbres Observatory (LCO) Network

Astronomy Research Seminar | Advisor: Dr. Rachel Freed

2025 NASA JET PROPULSION LABORATORY (NASA-JPL)

NASA Juno Mission

Analysis and Archiving of Near- and Mid-Infrared Observations of Outer Planet Atmospheres

Planetary and Exoplanetary Atmospheres Group

Student Independent Research Internship (SIRI) | Advisor: Dr. Glenn S. Orton

2025 CALIFORNIA INSTITUTE OF TECHNOLOGY (CALTECH)

| *NASA-JPL Lunar Trailblazer Mission*

2024 Planetary Sciences Department

Research Assistant, Science Communication Intern | Advisor: Dr. Bethany Ehlmann

2025 ORANGE COAST COLLEGE

Compact Objects and Gravitational Waves | Independent Study | Advisor: Prof. Jessica Asbell

ACADEMIC & WORKSHOP EXPERIENCES

2025 NASA AI/ML SCIENCE AND TECHNOLOGY INTEREST GROUP

PyTorch and Auto Differentiation Lecture

2025 COMPUTATIONAL RESEARCH ACCESS NETWORK | BRYN MAWR | NSF

Computational Research Methods for Plasma Physics & Sciences Virtual Seminar

Magnetohydrodynamics with FLASH, Monte Carlo Method, & Particle-in-Cell Tracks

2024 UNIVERSITY OF ILLINOIS | GRAVITY THEORY TRUST

EFTs, Gravity, and Fluid Dynamics: Progress, Challenges, and Emerging Opportunities Workshop

2024 CALTECH | NASA EXOPLANET SCIENCE INSTITUTE (NEXSCI)

Sagan Exoplanet Summer Workshop

Advances in Direct Imaging: From Young Jupiters to Habitable Earths

2024 MIAMI UNIVERSITY | NSF | EARTHSOPE CONSORTIUM

Seismology Skill Building Workshop

2024 QUEEN'S UNIVERSITY | CERN

The 6th Summer Particle Astrophysics Workshop (EIEIOO)

2024 ARIZONA STATE UNIVERSITY | NASA

Lucy Student Pipeline Accelerator and Competency Enabler (L'SPACE) Academy

NASA Proposal Writing and Evaluation Experience (NPWEE)

2024 NASA

NASA Community College Aerospace Scholars (NCAS)

2003 FRED HUTCHINSON CANCER CENTER | UNIVERSITY OF WASHINGTON

HutchLab | High School Human Genome Project

OBSERVATION EXPERIENCES

2026 NASA Infrared Telescope Facility (IRTF)

Mauna Kea, HI

2025 12 nights; 3.2 m; 0.8 - 5.3 μ m (iSHELL, SPEX, & MIRS)

Observing Jupiter in Support of Juno Mission; Volatiles in Comet Comae

2026 LCOZ - Siding Spring Observatory

Coonabarabran, AU

2 nights; 0.35 m; 0.4 - 0.7 μ m (QHY600)

Astrometry of double stars

2026 Caltech Palomar Observatory - Hale Telescope

Palomar Mountain, CA

4 nights; 5.1 m; 1 - 2.5 μ m (PHARO)

Observing Jupiter in Support of Juno Mission (NASA-JPL)

2025 LCO - Teide Observatory

Tenerife, ES

2 nights; 0.35 m; 0.4 - 0.7 μ m (QHY600)

Astrometry of double stars

2025	LCO - South African Astronomical Observatory <i>1 nights; 0.35 m; 0.4 - 0.7 μm (QHY600)</i> <i>Astrometry of double stars</i>	<i>Sutherland, ZA</i>
2025	Lockheed Martin Ignite Ground System (IGS) <i>4 nights; 8-9 kHz</i> <i>Monitoring for Lunar Trailblazer (LTB) Signal</i>	<i>Waterton, CO</i>
2025	Fukui University of Technology (FUT) Awara Space Center <i>2 nights; 8-9 kHz</i> <i>Monitoring for Lunar Trailblazer (LTB) Signal</i>	<i>Fukai, JP</i>
2025	Australia Telescope National Facility <i>6 nights; 8-9 kHz (Mopra)</i> <i>Monitoring for Lunar Trailblazer (LTB) Signal</i>	<i>New South Wales, AU</i>

TEACHING & TUTORING EXPERIENCES

2026	EARTHSCOPE CONSORTIUM MIAMI UNIVERSITY <i>August</i> <i>Education Intern</i> <i>May</i> <i>Seismology Skill Building Workshop</i> <i>Funded by NSF National Geophysical Facility (NGF)</i>	<i>Miami, OH</i>
2025	EARTHSCOPE CONSORTIUM MIAMI UNIVERSITY <i>September</i> <i>Teaching Assistant</i> <i>May</i> <i>Seismology Skill Building Workshop</i>	<i>Miami, OH</i>
2025	ORANGE COAST COLLEGE PLANETARIUM <i>May</i> <i>Presenter</i>	<i>Costa Mesa, CA</i>
2024		
2018	UNIVERSITY OF WASHINGTON <i>May</i> <i>Adjunct Instructor</i> <i>January</i> <i>Introductory Web Development (Extension)</i>	<i>Seattle, WA</i>

ENGINEERING EXPERIENCES

2023	CONTROLZEE, INC <i>April</i> <i>Senior Software Engineer</i>	<i>San Jose, CA</i>
2022		
2021	APPLE, INC <i>November</i> <i>Principal Software Engineer, Global Security Tools</i>	<i>Cupertino, CA</i>
2020		

2020	WEBFLOW	San Francisco, CA
April	Senior Software Engineer, Design System & Core Engine	
2019		
April		
2019	STARBUCKS COFFEE COMPANY	Seattle, WA
April	Lead Software Engineer, Web & Mobile	
2017		
November		
2017	WORLD WIDE TECHNOLOGY	St. Louis, MO
August	Senior Software Engineer	
2016		
August		
2016	ACTIVISION BLIZZARD	Irvine, CA
August	Software Engineer, Web & Mobile	
2015		
August		
2015	USA TODAY	McLean, VA
August	Senior Developer, New and Emerging Technologies	
2011		
November		
2011	NEUTRINO, INC	Seattle, WA
November	Front End Developer	
2008		
January		
2008	LUXURYREALESTATE.COM	Seattle, WA
January	Front End Developer	
2007		
May		

VOLUNTEER EXPERIENCES

2023	LABOR TECH RESEARCH NETWORK	St. Louis, MO
	Board Member	
present		
2021	COWORKER SOLIDARITY FUND	Annapolis, MD
October	Committee Member	
present		
2025	MESA STEM DAY	Newport Beach, CA
April	Workshop Assistant	

2024 <i>December</i>	MESA CAMPUS COUNCIL <i>Panelist</i>	<i>Newport Beach, CA</i>
2024 <i>November</i>	DEMOCRATIC NATIONAL COMMITTEE <i>Phone Bank Operator</i>	<i>Washington, DC</i>
2024 <i>October</i>	NEWPORT-MESA COLLEGE & CAREER FAIR <i>Booth Operator for Coastline College & MESA</i>	<i>Costa Mesa, CA</i>
2024 <i>September</i>	EXPLORE CALTECH <i>Booth Operator for Pasadena City College Physics & Astronomy</i>	<i>Pasadena CA</i>
2024 <i>May</i>	BOLSA CHICA CONSERVANCY <i>Invasive Species Removal</i>	<i>Huntington Beach, CA</i>
2024 <i>May</i>	SAN DIEGO COAST KEEPER <i>Water Quality Testing</i>	<i>San Diego, CA</i>
2023 <i>August</i>	SURFRIDER FOUNDATION <i>Beach Cleanup</i>	<i>San Clemente, CA</i>
2015 <i>August</i>	ORANGE COUNTY PROBATION DEPARTMENT <i>At-risk youth mentor (OC GRIP)</i>	<i>Santa Ana, CA</i>
2015 <i>August</i>	MINDMAKERS <i>Youth mentor & web developer</i>	<i>Pasadena, CA</i>

AWARDS & SCHOLARSHIPS

2026	AMERICAN INSTITUTE OF PHYSICS <i>TEAM-UP Together Research Fellow</i>	<i>Silver Springs, MD</i>
2026	WELLESLEY COLLEGE <i>Elizabeth Kaiser Davis Scholar</i>	<i>Wellesley, MA</i>
2026	NATIONAL OCEANIC & ATMOSPHERIC ADMINISTRATION <i>Ernest F. Hollings Scholar</i>	<i>Silver Springs, MD</i>
2026	JACK KENT COOKE FOUNDATION <i>Transfer Scholar Semifinalist</i>	<i>Lansdowne, VA</i>
2026 2025	SANTA BARBARA CITY COLLEGE <i>President's Honor Roll</i>	<i>Santa Barbara, CA</i>
2025	WOMEN'S INDEPENDENCE SCHOLARSHIP PROGRAM <i>Doris Buffett Independence Scholar</i>	<i>Wilmington, NC</i>
2025	AMERICAN INSTITUTE OF PHYSICS <i>TEAM-UP Together Scholar</i>	<i>College Park, MD</i>
2025	ORANGE COAST COLLEGE <i>Garrison Fellow</i>	<i>Costa Mesa, CA</i>
2025	COMMUNITY COLLEGE LEAGUE OF CALIFORNIA <i>All-California Academic Award, First Team</i>	<i>Sacramento, CA</i>

2025	COASTLINE COLLEGE FOUNDATION <i>Associated Student Government Career & Technical Merit Scholarship</i>	<i>Fountain Valley, CA</i>
2025	COASTLINE COLLEGE FOUNDATION <i>Herman Tate Endowment</i>	<i>Fountain Valley, CA</i>
2025	JEANNETTE RANKIN FOUNDATION	<i>Athens, GA</i>
2024	<i>Jeannette Rankin National Scholar</i>	
2025	NATIONAL SCIENCE FOUNDATION	<i>Alexandria, VA</i>
2024	<i>S-STEM Scholar (Coastline College)</i>	
2024	ORANGE COAST COLLEGE FOUNDATION <i>Peter Christian Hernandez Memorial Scholarship Award</i>	<i>Costa Mesa, CA</i>
2022	HACKER NOON <i>Developer Defying Patriarchy (While Being an Awesome Developer) (Runner-up)</i>	<i>Edwards, CO</i>

PUBLICATIONS

- Scarlett, C. (2026). Finding Physicists: The vital role of Community Colleges. *The SPS Observer*, 60(1).
<https://students.aip.org/observer/volume-60-number-1>
- Scarlett, C. (2025). Lunar Trailblazer Interactive Model [Computer software]. *California Institute of Technology*.
<https://trailblazer.caltech.edu/spacecraft/index.html>
- Scarlett, C. (2025). Water into Doom: Balancing AI with Climate Action & Environmental Costs at the Water-Energy Nexus. *The Honors Transfer Council of California Building Bridges Research Conference*.
- Scarlett, C. (2025). The Climate and Tornadoes are Changing. Can Understanding Climate-Driven Chaos Save Lives?. *Academia*.
https://www.academia.edu/150281326/The_Climate_and_Tornadoes_are_Changing_Can_Understanding_Climate_Driven_Chaos_Save_Lives
- Scarlett, C. (2025). Bullshit at the Agora: On the Dangers of a 'Marketplace of Ideas'. *Academia*.
https://www.academia.edu/146050292/Bullshit_at_the_Agora_On_the_Dangers_of_a_Marketplace_of_Ideas
- Scarlett, C. (2025). No Sleep 'Til Payday: Class, Work, and the Price of Survival. *Academia*.
https://www.academia.edu/145993371/No_Sleep_Til_Payday_Class_Work_and_the_Price_of_Survival
- Scarlett, C. (2025). Emmy Noether: A Woman on the Edge of [Space]Time. *Academia*.
https://www.academia.edu/144259985/Emmy_Noether_A_Woman_on_the_Edge_of_Space_Time
- Scarlett, C. (2024). Man Made: Technology, Pygmalion displacement, and the patriarchal idealization of women in science fiction. *Academia*.
https://www.academia.edu/145993413/Man_Made_Technology_Pygmalion_displacement_and_the_patriarchal_idealization_of_women_in_science_fiction
- Scarlett, C. (2024). Remote seismology correlation and Mount St. Helens volcanology. *Seismology Skill Building Workshop*.
https://nbviewer.org/urls/www.iris.edu/hq/files/short_courses/2020/ssb/mt-rainier-seismicity.ipynb

- Scarlett, C. (2024, July 29). *A case of the blues: Is Overexposure to Artificial Light Causing a Mental Health Crisis?*. Esri. <https://storymaps.arcgis.com/stories/e236a32458bb4b7db87171f6a0305d8a>
- Carlson, G., Roginsky, J., & Scarlett, C. (2022, February 6). Washington can be a leader on protecting sexual harassment, assault survivors. *The Olympian*.
<https://www.theolympian.com/opinion/op-ed/article258066068.html>
- Scarlett, C. (2019, March 12). Creating a Spotify-powered app using Nuxt.js. *Smashing Magazine*.
<https://www.smashingmagazine.com/2019/03/spotify-app-vue-nuxt-javascript/>
- Scarlett, C. (2008). sIFRvault [Computer software].